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SHORT COMMUNICATION

WILEY

What can surface wind observations tell us about interannual variation in wind energy output?

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Abstract

The past decade of wind power growth was supported by capacity factor improvements and associated cost reductions. But are higher capacity factors a technology success story or, as suggested by recent research, has the influence of technology been overstated by ignoring positive surface wind speed trends? The answer could influence estimates of wind energy's cost and even future deployment rates. We find that US surface wind speed observations imply a 2.6% improvement in capacity factors from 2010 to 2019. Yet newer vintages of wind plants have recorded capacity factors that are ∼25% larger than plants built close to 2010. It follows that technological factors and improved site quality, not higher wind speeds, drove most of the improvement in capacity factors. Additionally, we match hundreds of meteorological stations to nearby (<25 km) wind plants and compare annual estimated generation, based on a function of surface wind speed observations, to annual recorded generation. Researchers rely on this publicly available surface data because measurements co-located with wind plants are generally considered proprietary. Our analysis addresses a research gap: interannual variation in observed surface wind speeds is rarely compared to observed data at wind plant locations and turbine heights. We find that despite its common use for this purpose, generation estimates based on publicly available surface observational data provide a poor proxy for interannual variability in recorded wind generation. These findings suggest that caution is generally needed when researchers use surface wind speed measurements to investigate long-term wind energy trends.

KEYWORDS
capacity factor, surface wind observations

1 | INTRODUCTION

Between 2010 and 2019, the average capacity factor of the full US wind fleet increased by 13%, from 0.307 to 0.347, with capacity factors from new plants averaging >0.4 in recent years.[1] This increase in capacity factor is a key driver of record-low levelized costs of new wind (and record levels of new wind deployment). It is thus important to understand what has caused the increase in capacity factor. While it is commonly assumed that improvements in wind turbine technology are primarily responsible for this increase, there is also some debate as to whether wind speeds have increased during this period, contributing to the observed capacity factor increases. If it is true that capacity factor increases were, to a large degree, driven by increasing wind speeds, that would suggest that the commonly assumed influence of technology change on capacity factor is overstated and that some of the recent gains to capacity factors might therefore be lost if wind speeds were to revert to levels seen a decade ago. This result could negatively affect wind energy cost assumptions and deployment, potentially slowing carbon mitigation efforts.

In particular, recent research has suggested that the increase in wind plant capacity factors is overstated[2] or cannot be attributed to technology change as much as commonly assumed.[3] Zeng et al.,[3] for example, show that global surface wind speeds, which had been trending downward between 1980 and 2010, reversed course in 2010 and trended upward through 2017. They argue that this increase in surface wind speeds was also likely experienced by wind plants, accounting for roughly half the increase in average wind capacity factors in the United States over this period.

In this short communication, we use plant-level generation records and surface wind speed observations to investigate two questions: one, is there evidence that increased wind speeds account for a substantial portion of the capacity factor growth observed over the last decade? Two, how do interannual trends in surface wind speed observations relate to interannual trends in recorded capacity factor?

Why do researchers use surface wind speed observations to infer changes in the wind resource available to wind plants? Simply put, the two most logical alternatives present their own challenges. Wind speed observations that are measured at altitude (modern tower heights average ∼90 m) at actual wind plant sites are typically held as private data and not made publicly available.[4] Meanwhile, various reanalysis datasets that are based on global meteorological models are publicly available and do offer complete temporal and geographical coverage (and at various altitudes), but their modeled long-term trends in surface wind speeds can differ from one another, as well as from observed trends.[5–8] As such, researchers hoping to maintain a focus on observed rather than modeled wind trends must make use of publicly available data from surface meteorological stations.

A common approach to estimate potential wind generation from surface wind speed observations involves two steps. First, surface wind speed is extrapolated to the tower height (or "hub height") using a power law. Second, a turbine manufacturer's power curve is used to estimate the likely generation as a function of the wind speed. This approach is sometimes used with surface observations that are co-located with wind plants but is also commonly applied at a regional-level, as in previous studies.[3,6] That is, the process is applied to surface observations across a region and used to estimate regional trends in wind generation. Though this practice is fairly common, it is not without controversy. In particular, there are questions about whether this process is appropriate given evidence of a disconnect between surface and hub-height wind speeds (e.g., rotor tips can extend beyond the atmospheric surface layer) and the often sizeable horizontal distance between observation stations and wind plants.[9–11]

Our analysis addresses a research gap: to date, empirical analyses that compare interannual wind speed variation at surface meteorological stations to data at active wind plants (at hub-height) have been limited due to the challenges listed above. Our use of generation records allows us to address this research gap with data produced and recorded by wind plants themselves. We briefly explain our methods in Section 2. We present results related to region-wide trends in Sections 3 and 4. We present analysis of paired observation stations and wind plants in Section 5. Our concluding discussion (Section 6) summarizes how best to make use of surface observations for describing interannual variability in potential wind generation and assesses the relative importance of technology change versus wind speed trends as drivers of capacity factor improvements over the last decade.

2 | METHODS

2.1 | HadISD global sub-daily dataset

The HadISD dataset, hosted at the Met Office Hadley Centre, contains meteorological measurements at surface weather stations around the world.[12,13] We analyzed data from 2010 to 2019. We used version 3.1.2.202108p and downloaded sub-daily wind speed observations (10 m above ground). We excluded stations outside of the United States.

We classified how close observation stations were to wind plants by finding if the centroid of a wind plant was located within a certain radius of the station (analyzing whether plants were within 10, 15, 25, 50, or 100 km of a station). Geocoded wind plant centroids were based on the United States Wind Turbine Database.[14] Figure 1 shows the location of HadISD stations and their proximity to wind plants.

2.2 | Calculated capacity factors

HadISD observations are sub-daily, meaning that in many cases, they are hourly but can range to a 6-h frequency. In order to develop estimates of capacity factors based directly on observed wind speeds, we created consistent hourly wind speed records. We roughly followed the methods

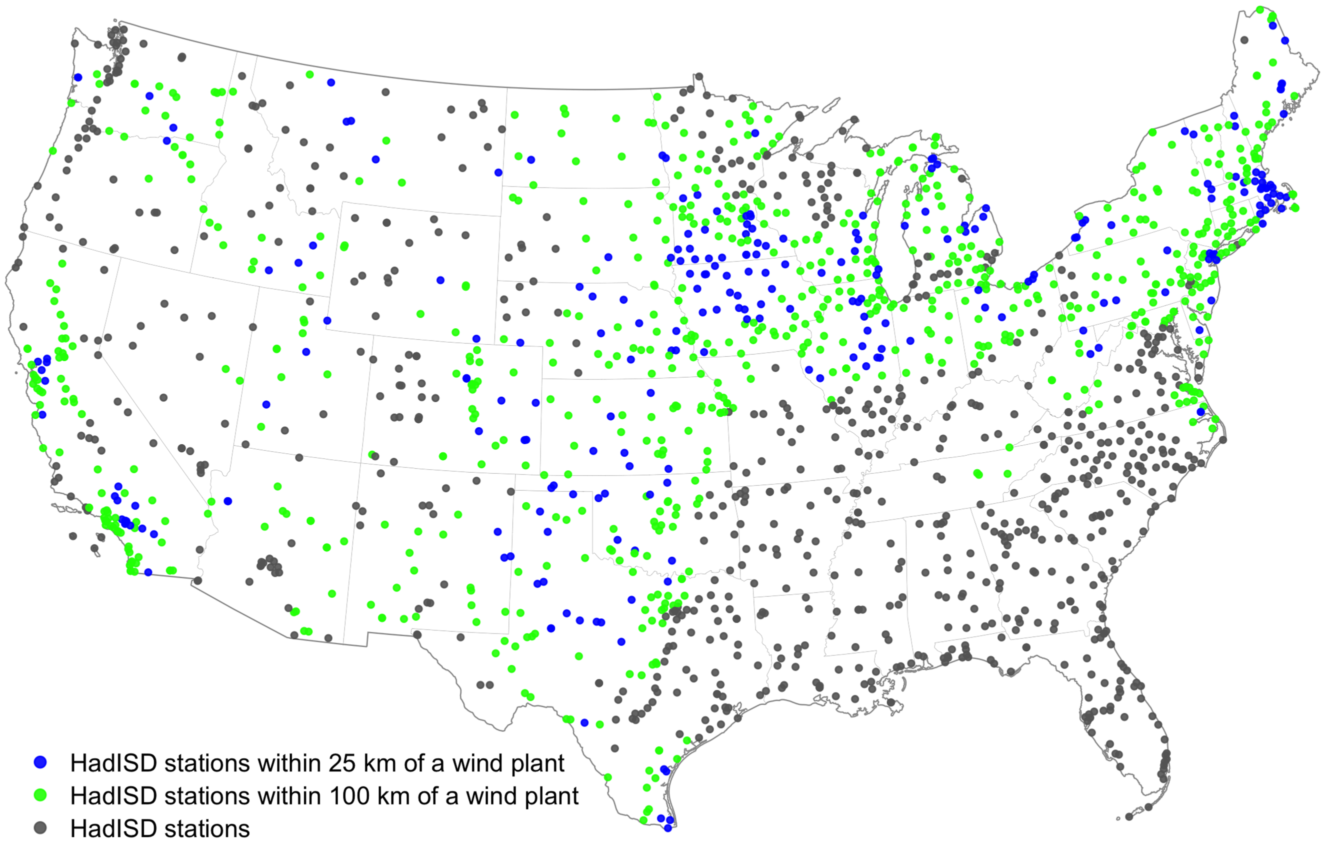

FIGURE 1 Weather stations included with HadISD. The stations are color-coded by proximity to wind plants

of Zeng et al.[3] and interpolated wind speeds between hours (where hourly data were missing) and then extrapolated wind speeds to hub-height and used a manufacturer's power curve to estimate potential wind generation as a function of observed wind speeds.

We linearly interpolated missing wind speeds for up to a 6-h period; any period of missing values longer than 6 h was left as missing. Surface wind speeds were then extrapolated from 10 m up to two different hub heights, 90 and 140 m, using the simple power law. The power law, following Zeng et al.,[3] is shown in Equation 1.

$$u_{tb} = u\left(\frac{Z_{tb}}{Z_s}\right)^a \tag{1}$$

where a is a constant of 1/7, u_{tb} is the wind speed at the turbine height, u is the observed surface wind speed, Z_{tb} is the turbine height, and Z_s is the height of the observation station (10 m).

A power curve for a GE-2.5-120 turbine was used to transform the hub-height wind speeds into wind generation. This power curve begins generating at 3 m/s and reaches rated capacity at 12 m/s, and follows the typical s-shape curve. We normalized the power curve to find an hourly capacity factor. We refer to this quantity as the “calculated capacity factor.” This is in contrast to the “recorded” generation or capacity factors, which represent actual reported data from existing wind plants.

Annual average calculated capacity factors are determined using a two-step process designed to minimize the influence of missing data.[1] The average capacity factor at each hour of the day is found for each month (i.e., 24*12 = 288 values).[2] A weighted average of these values is calculated, with weights based on each hour's fraction of total hours in the year, accounting for the differing number of days in each month and leap years. If no data is missing, this approach results in a simple average of capacity factors across all hours in the year. If data are missing, this process ensures that the diurnal cycle in each month is appropriately weighted and that hours with a greater portion of data available are not overly weighted. For example, we do not want to overweight nighttime observations because there are fewer missing values during the nighttime. Finally, an annual average is calculated only if all of the 288 hourly values are based on >70% of possible data points for each hours. This is a strict criteria: if an observation station has missing data for 10 days of a single month, the annual average for that station and year would not be included. This strict criteria ensure that interannual variation in annual average wind speed is not due to sampling differences and missing data.

For the 2010–2019 trends explored in Section 3, stations are only included if they have at least 8 years of data over the time period; this reduces the available set of stations from 1749 to 1409, though many of those stations are far from wind plants, as shown in Figure 1.

2.3 | Wind plant generation records

Wind plant generation records were based on the data developed for Hamilton et al.[15] These data can be accessed at https://emp.lbl.gov/publications/how-does-wind-project-performance. The data were updated through 2019. To briefly summarize the methods here, generation by plant is based on Energy Information Administration (EIA) Form 923.[16] The EIA form provides monthly generation records for all utility-scale wind plants in the United States. Reported generation was adjusted upwards for curtailment using reported curtailment estimates at the regional level. This curtailment was assigned to individual plants based on plant characteristics and local nodal pricing (based on the assumption that plants in locations with low and negative prices had higher levels of curtailment). Additional details can be found in Hamilton et al.[15]

3 | THE IMPORTANCE OF PLANT DESIGN AND TECHNOLOGY VERSUS WIND SPEED TRENDS

We compare trends among various subsets of observation stations based on their distance to wind plants (Figure 2). Though stations close to wind plants record consistently higher wind speeds and calculated capacity factors (to be expected, given that wind plant developers seek the windiest locations for their plants, all else equal), the interannual trends in wind speeds and calculated capacity factors are essentially the same for all groups. Thus, we can conclude that, across the contiguous United States, the region-wide average interannual trends in surface wind speed and calculated capacity factors are robust to site selection (provided the groups are of sufficient size).

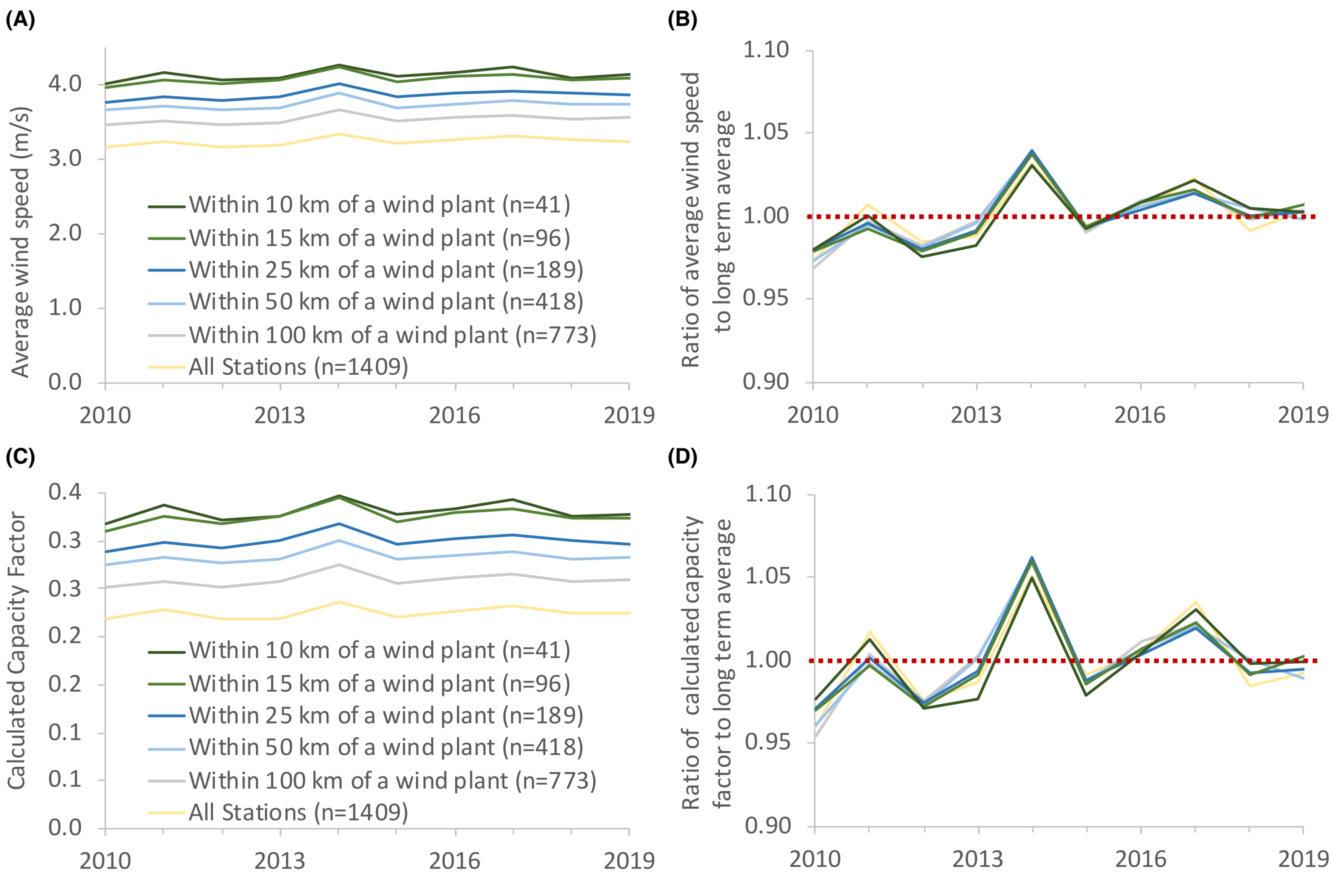

FIGURE 2 Surface observation stations that are located close to wind plants have higher average wind speeds, but similar interannual variation. (A) annual average wind speed; (B) annual average wind speeds normalized by the long term average, 2010–2019; (C) annual average calculated capacity factor; (D) annual average calculated capacity factor normalized by the long term average

Calculated capacity factors (Figure 2C,D) have similar interannual trends to surface wind speeds (Figure 2A,B), but the interannual variability of calculated capacity factors is amplified (due to the extrapolation to hub-height and application of a power curve). Calculated capacity factors were generally lower in 2010–2012 than in 2017–2019. But the difference is small: using the 25 km distance cutoff, average calculated capacity factors in 2017–2019 averaged 0.30, just 2.6% larger than the 0.29 average calculated capacity factor from 2010 to 2012. Note that this percentage increase in calculated capacity factor is roughly equal across all observation station groupings.

In contrast to these mild trends in surface wind speed and calculated capacity factors, we see dramatic upward trends in actual recorded capacity factors associated with plant vintage over this same period (Figure 3). Generally, plants built in 2014 and later had capacity factors (in 2016–2019) that were considerably higher than all previous vintages. A more specific comparison: second-year capacity factors for plants built in 2009–2011 averaged 0.32 while second-year capacity factors for the newest plants in our study (built in 2015–2017) averaged 0.40, a 25% increase. The 25% increase to second-year capacity factors for newer vintages is an order of magnitude greater than the 2.6% increase to calculated capacity factors implied by observed surface wind speeds alone, which implies that technological factors/plant design and the quality of chosen sites, not region-wide trends in wind speeds, were the dominant driver of improved capacity factors over the last decade.

The quality (i.e., estimated long-term capacity factor) of chosen sites for new wind plants improved over the past decade. Site quality estimates were made while holding technology assumptions and the historical window of evaluation constant, meaning they were independent of the technology and wind speed trends. The site quality of plants built in 2015–2017 was 6% improved over the site quality of plants built in 2009–2011.[1] This implies that, given the same turbine, the average output would be 6% higher at new plant locations in the latter period compared to the former period. Site selection is a complicated process and is not independent of technology improvement: most importantly, improved technology allows for a broader set of sites to be even considered for new plant development. We do not try to further separate the influence of site quality versus technology and plant design.

4 | REGION-WIDE INTERANNUAL VARIABILITY: CALCULATED CAPACITY FACTORS VERSUS RECORDED CAPACITY FACTORS

Here, we again examine the region-wide annual averages from Section 3, but we isolate two separate 3-year periods in order to highlight similarities and differences between interannual variation in region-wide calculated capacity factors and recorded capacity factors. Interannual variability in region-wide recorded capacity factors only loosely followed trends in region-wide calculated capacity factors. For example, Figure 4A shows that while recorded capacity factors and calculated capacity factors both peaked in 2014, the 6% increase above 2013 for calculated capacity factor was roughly double the 3% increase in recorded capacity factor. Another example, Figure 4B shows that, compared to 2016, 2017 recorded capacity factor declined while calculated capacity factor increased; 2018 calculated capacity factors declined from 2017 and recorded capacity factors declined similarly, but the older vintages' recorded capacity factors stayed flat. The main point here is that region-wide average calculated capacity factor is an imprecise proxy for interannual trends in recorded generation, which implies that surface wind speed observations are an imprecise proxy for interannual trends in elevated wind speed at wind plants. In the next section, we examine paired observation station and wind plant records to gain more insight into the usefulness of these surface wind speed observations for estimation of wind plant generation.

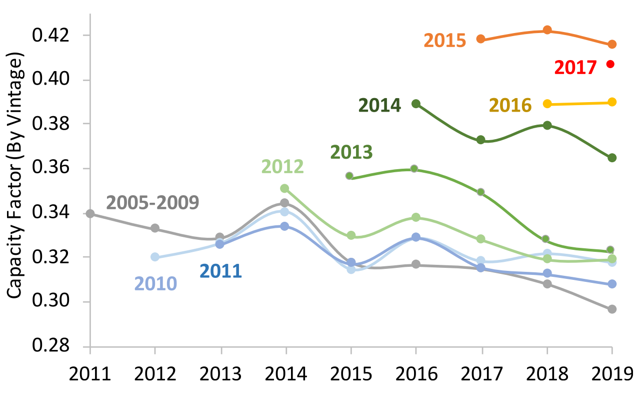

FIGURE 3 Newer plants have capacity factors that are much higher (in the same calendar years) than older plants, and the difference between vintages is larger than the interannual variability. Note, the first full year of production is excluded from each vintage to account for staged installation and "teething" issues

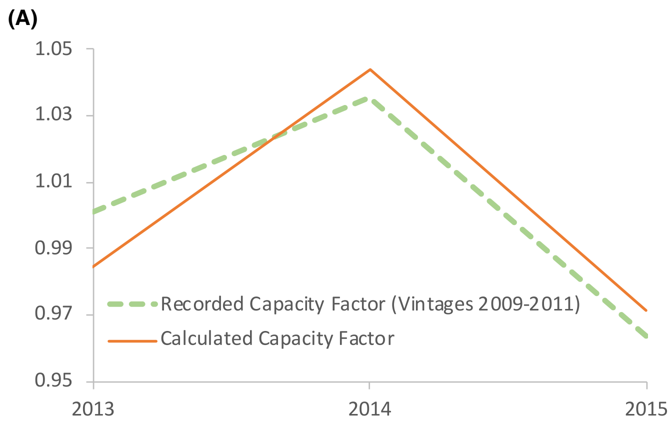

FIGURE 4 Comparison of region-wide, three-year trends in vintage-specific recorded capacity factors and calculated capacity factors for years (A) 2013–2015 and (B) 2016–2018. Region-wide calculated capacity factors are an imprecise proxy for recorded capacity factors as calculated capacity factors overestimate improvement between 2013 and 2014 and suggest an increase, rather than recorded decline, between 2016 and 2017. All values were normalized by their average values over each 3-year window, and calculated capacity factors were averaged across all stations within 25 km of a wind plant

5 | COMPARISON OF INTERANNUAL VARIATION AT METEOROLOGICAL STATIONS AND NEARBY PLANTS

In this section, we compare interannual trends in calculated capacity factors at individual observation stations to interannual trends in recorded generation at matched, nearby wind plants. We examine a very specific metric: the normalized change between year X and year X-1. For example, if the annual calculated capacity factor at an observation station was 0.33 in 2013 and 0.30 in 2012, we would give that station a value of 1.1 (i.e., 0.33/0.30) for year 2013. We calculate a similar value for the normalized change in recorded capacity factor at the matched, nearby wind plant, and then compare the two values.

We focus on differences between adjacent years (rather than on longer-term trends) because many observation stations maintain records for only a few years or have multiple years with missing data. By focusing on differences between single years, any observation station that contains at least two consecutive years of data can be included. Similarly, the fleet of wind plants has grown continuously, and this approach allows us to include wind plants built in the latter half of the study period. An additional reason to focus on adjacent single years is that plant performance declines with age. Over 2 years, this performance decline is typically small compared to interannual variation, but over many years, could be similar in magnitude. Thus, the focus on differences between adjacent years allows us to mostly ignore performance decline with age. Finally, note that we removed the first year of plant operation to avoid issues related to phased construction and general "teething issues," and we also removed years in which recorded generation changed by more than 30% from the prior year, as we assume that changes of this magnitude most often reflect maintenance or other operational change that is not related to wind speed variation.

We find that interannual changes in recorded capacity factor are *not* closely correlated with interannual changes to calculated capacity factor (see Figure 5). Linear regression of the two quantities yields a coefficient of determination of 0.14 and a slope of 0.47. This indicates both a lack of correlation and that only about half of any increase (or decrease) in the calculated capacity factors is reflected in recorded capacity factors.

The above discussion uses a distance cutoff of 25 km and includes an extrapolation of surface wind speeds to 90 m above ground for input into for the calculated capacity factor. These are reasonable choices as the hub height of most modern US wind plants is ~90 m and many wind plants span horizontal lengths of 10 to 20 km. However, we ran the same analysis as presented in Figure 5 but varied both the distance cutoff and the assumed hub height. We note our sample size changes with our choice of cutoff distance—a smaller distance leads to fewer wind plants and observation station pairs.

Figure 6 shows statistics for the combination of cutoff distances and hub heights (the slope is shown in panel A and the coefficient of determination is shown in panel B). Slopes range from 0.24 to 0.54 indicating that in no grouping did recorded capacity factors fully reflect the year-to-year change in calculated capacity factors. We note that having a strict distance cutoff of 15 or 10 km reduced coefficients of determination to close to zero. This loss of explanatory power was likely due to the reduction in sample size (251 points with a 10 km distance requirement versus 2360 and 8769 with 25 and 50 km requirements, respectively). Interestingly, the coefficient of determination also declined at the 100 km distance level as well, despite continued increase to sample size, indicating the correlation declines as the distance between observation station and plant grows beyond 50 km. All the slopes have a significance level of 0.99. Finally, all the intercepts (not shown) are equal to a value such that no change in observed wind speed is associated with a ~1% decline to recorded capacity factor, a result roughly consistent with literature on plant performance decline with age.[15,17,18]

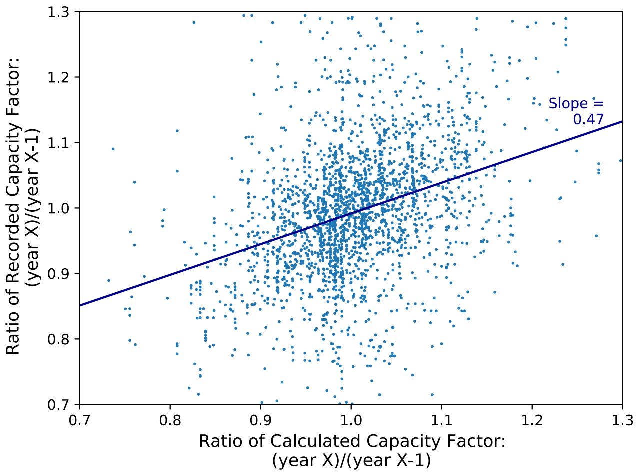

FIGURE 5 Comparison of year-to-year changes in recorded capacity factor versus calculated capacity factor. Each point reflects data from a paired observation station and plant and reflects the change in each variable between two adjacent years. The slope of a simple linear regression of the form y = mx + b is included

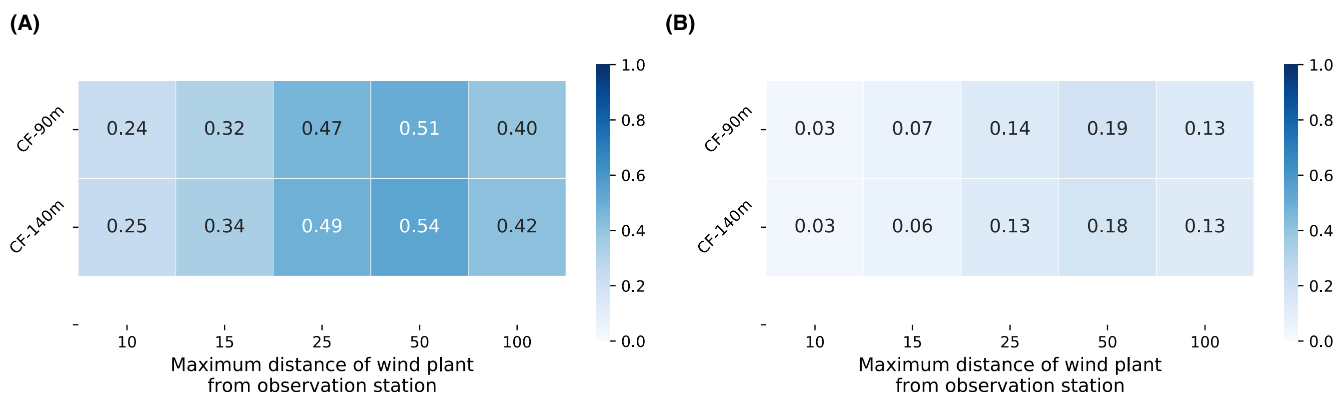

FIGURE 6 Summary of (A) the slope and (B) coefficient of determination, of the simple linear regression presented in Figure 5, but across a range of different pairings of meteorological stations and wind plants (varying the allowed distance for a pairing) and with the independent variable set to calculated capacity factor (CF) at 90 or 140 m above ground. Note that all slopes presented are significant at a 99% confidence level

6 | CONCLUSION

Between the beginning and end of last decade, the average recorded capacity factor of the fleet of wind plants in the United States rose substantially. There has been debate in the literature about how much of this capacity factor increase can be attributed to technological changes versus underlying trends in wind speeds. Here, we found that the average, second-year, capacity factor of the newest plants in our study (built from 2015 to 2017) was 0.40, 25% larger than the 0.32 average second-year capacity factor of plants built from 2009 to 2011. In contrast, calculated capacity factors (which are a function of observed surface wind speeds) increased by only 2.6% over this same time period. This suggests that technological changes and improved site quality are by far the dominant factor in the last decade's improvements in observed capacity factor.

The term "technology change" applies to a number of different aspects of wind plants. Important plant design changes have been carefully documented in existing literature. Most prominently, newer plants have larger blades relative to turbine capacity. These larger blades allow turbines to generate more power at lower wind speeds and thus increase wind plant capacity factors. Additionally, tower heights have mildly increased over the past decade, further improving capacity factors. The quality of chosen sites for new development can also influence capacity factors of new plants. We found that plants, built from 2015 to 2017, had site quality that was improved by 6% from plants built in 2009–2011. Of course, site selection depends to some extent on available technology and plant design options, and we did not try to further separate the importance of site selection from technology changes.

A second conclusion from this paper is that generation estimates based on the network of publicly available surface wind speed observations provides only a weak proxy of interannual variation in actual recorded generation. Specifically, we found that calculated capacity factors derived from surface wind speed data from meteorological surface stations can explain only 19% of the annual changes in recorded generation at nearby wind plants (based on the maximum coefficient of determination from Figure 6). Furthermore, we found that the slope was ∼0.5, indicating, for example, that a 5% year-to-year change in calculated capacity factors was associated with only 2.5% change in recorded capacity factors.

This second conclusion is important because it contests the common assumption that interannual trends in wind speeds found at the network of surface meteorological stations are a good proxy for interannual trends of wind speeds at wind plants. There are two potential reasons that wind trends at observation sites may differ from wind trends at wind plant locations and at wind hub-heights. One is that even a 10 km distance between observation site and wind plant might dull the correlation of interannual trends. Second is that phenomena outside the surface boundary layer (such as low level jets) influence hub height wind speed trends but would not be reflected in observations of surface winds.[9] We suggest that until the relationship between observed surface wind speeds and wind plant wind speeds is more fully understood, caution should be taken when using surface wind speed observations to draw conclusions about long-term wind energy potential.

Of course, the need for this caution should be applied to this very paper: even the modest 2.6% increase to calculated capacity factors over last decade is possibly an overestimate of the actual increase to recorded capacity factors that can be attributed to wind speed trends. If so, this would imply that technology was an even more important driver of improved recorded capacity factors.

There are some important limitations to our analysis. Our analysis examined wind plant and observation station pairs that are within 10 to 50 km of each other, and thus, we cannot comment on the relationship between surface wind speeds and potential wind energy output when an observation station is exactly co-located with a wind turbine. That said, many wind plants span horizontal dimensions that are similar in length to the distances we examined (i.e., 10 to 20 km), and thus, these results may have relevance to the use of a single surface monitor placed within a large wind plant. Another limitation of this study is that we only examined interannual variability; our results do not speak to monthly or hourly time-scales.

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PEER REVIEW

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DATA AVAILABILITY STATEMENT

Much of the data that support the findings of this study are publicly available. Additionally, data is available on request from the corresponding author though a small portion of the data is not publicly available due to commercial privacy restrictions.

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